

Solution Architecture

Date	10 June 2026
Team ID	xxxxxxx
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>[Presentation Layer]</i>	Provides an intuitive, responsive frontend web portal allowing users to seamlessly record raw speech explanations, view live visualization matrices, and query system scores.	<i>Stream lit, HTML, CSS, Python</i>
<i>[API Gateway]</i>	Manages active application loopbacks, handles session state initialization, routes structural text variables, and coordinates multi-threaded processing queues safely	<i>Python, Stream lit Session State, Async Processing, File Handling Modules</i>
<i>[Microservices]</i>	Executes acoustic feature isolating algorithms, generates text transcripts from audio streams, maps semantic vector distance matchings, and formats dynamic grading distributions.	<i>OpenAI Whisper, Sentence Transformers (SBERT), Transformers, PyTorch, Scikit-learn, Librosa, NumPy, Matplotlib, NLTK, Report Lab</i>

[Database]	Offers structured local staging directories to store incoming user audio segments temporarily while securely tracking output assessment summaries and generated PDFs.	Local File System, Python OS Module, Temporary Directories, PDF Storage
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